

(Please write your Enrolment No. immediately)

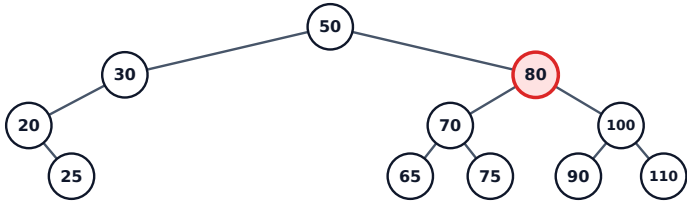
Enrolment No: _____

MID TERM EXAMINATION
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
B.Tech — 3rd Semester • September, 2025
Paper Code: CIC-209 • Subject: Data Structures

Time: 1½ Hours

Maximum Marks: 30

Note: Question 1 is compulsory. Attempt **any two** more questions from the remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Differentiate between an array and a linked list in terms of memory allocation, access time, and insertion/deletion efficiency.	[2]	CO1
1 (b)	Define a double-ended queue (Deque). What are its types (Input-restricted and Output-restricted Deque)?	[2]	CO1
1 (c)	Define Sparse Matrix. How can it be represented in memory (using 3-tuple coordinate format and linked lists)?	[2]	CO2
1 (d)	Differentiate between a binary tree and a full binary tree by taking a suitable example.	[2]	CO2
1 (e)	Write the overflow and underflow conditions of a circular queue implemented using an array of size N.	[2]	CO2
2 (a)	Consider 1-D arrays A(5:50), B(-5:10), and C(18): (i) Find the number of elements in each array. (ii) Suppose Base(A) = 300, w = 4 words/memory cell for array A. Find the memory address of A[15], A[35], and A[55]. OR Convert the following infix expression to postfix expression showing each stack step: $((A + B) - (\{C + D\} * E) / F) * G$	[5]	CO1
2 (b)	For the given Binary Search Tree (BST), what will be the resulting tree after deleting element 80 from this tree? (Show the replacement using in-order successor or in-order predecessor):  OR Write an algorithm/program to reverse a singly linked list.	[5]	CO2
3 (a)	Write an algorithm/program to insert (push) and delete (pop) elements from a stack using array representation. OR What is the limitation of a linear queue? Discuss the advantages of using a circular queue over a linear queue.	[5]	CO2
3 (b)	Write an algorithm/program to create a doubly linked list and traverse each node in both directions (forward and backward). OR Define a max-heap. Build a max-heap step-by-step for the following data: 18, 16, 22, 20, 15, 40 .	[5]	CO3